

made by Jessica Leo

Mobile Game Monetization Analytics

Why This Project?



Most monetization analysis stops at aggregate revenue — total IAP income, overall ARPU. That hides the real story: a small fraction of players usually drives most of the revenue, and treating every paying user the same leads to misallocated retention spend.



This project segments players by actual spend behavior (Whale/Dolphin/Minnow) and cross-validates every segment across two independent queries, rather than trusting a single aggregate number.

Dataset & Methodology



Dataset: 3,024 mobile game users, 8 months of transaction data (Kaggle)

Tools:

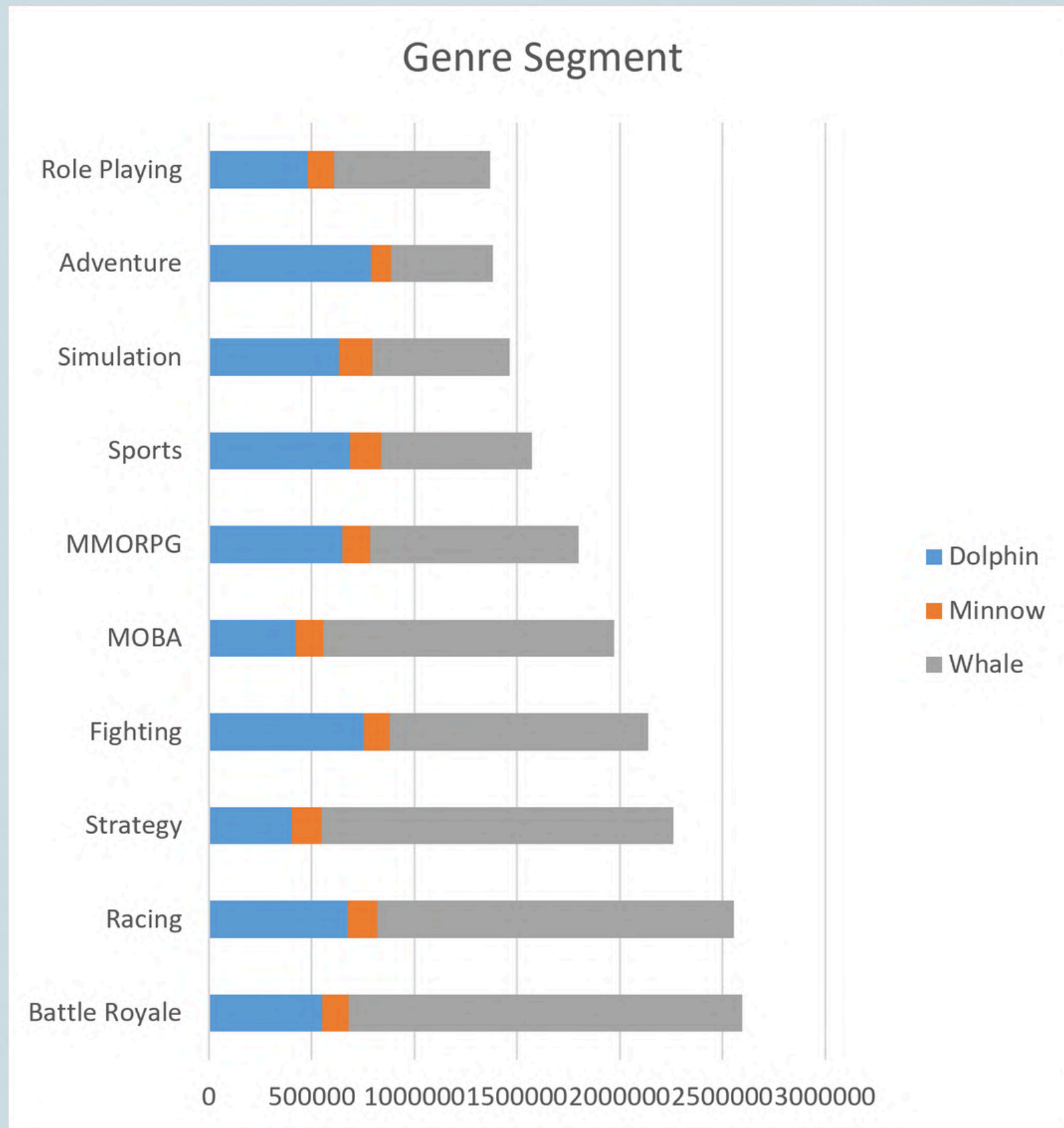
- SQLite (CTE-based segmentation)
- Excel (PivotTables & Charts)

Note: an initial $\geq \$100$ "Whale" threshold, carried over from a generic segmentation template, misclassified 91.8% of users as Whale once run against this dataset's actual spend range. Thresholds were recalibrated against the real distribution, and 130 zero-value transactions were excluded before finalizing segments.

Key Insight #1

A Small Minority Drives Most Revenue

2.2% of paying users (Whales) generate 58.3% of total revenue, while 84.1% of users (Minnows) contribute only 8.5%.

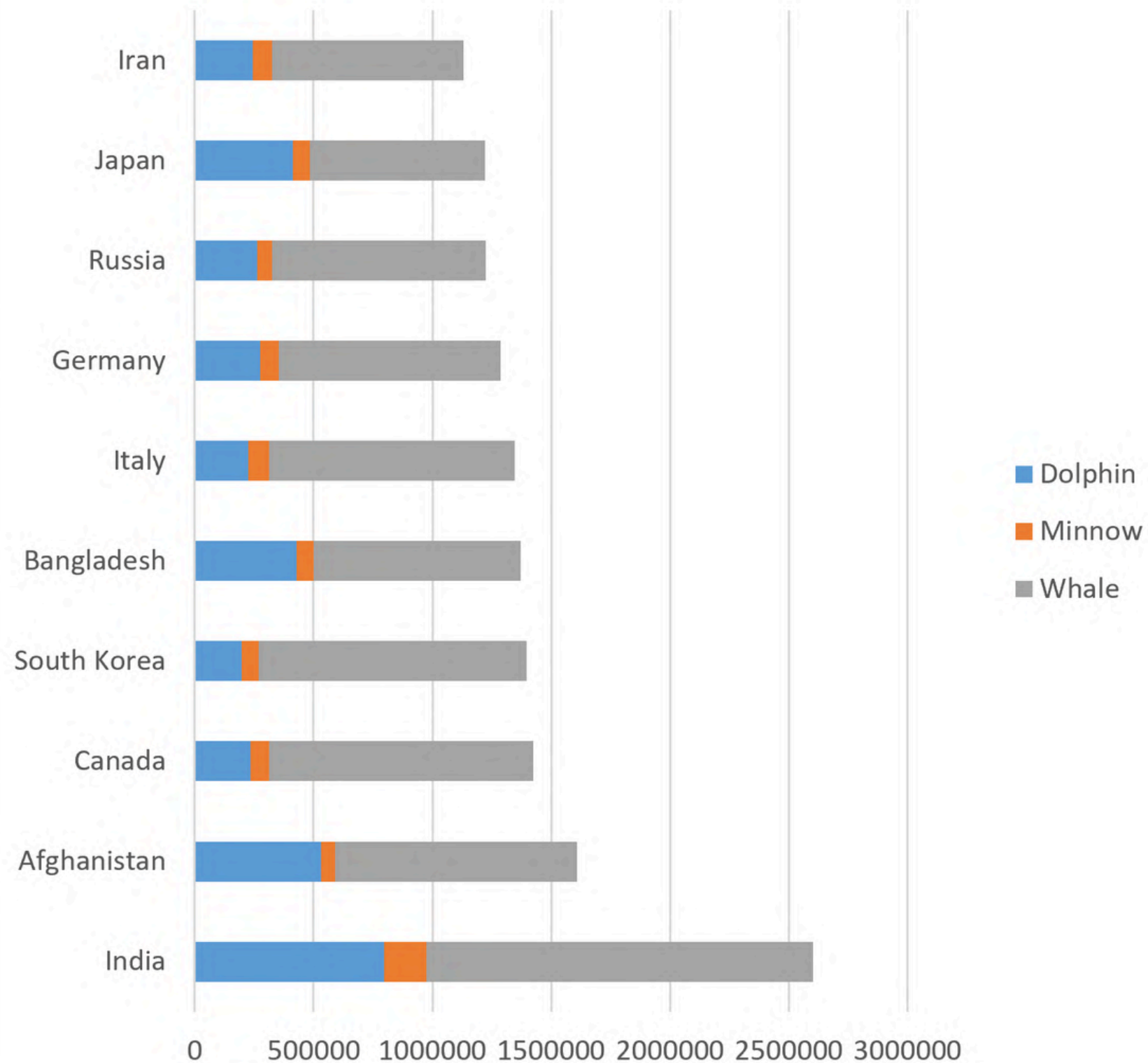


Key Insight #2

Volume and Value Live in Different Markets

India is the highest-volume market (237 players, \$2.6M revenue) but only mid-tier on ARPU (~\$10,970). Afghanistan has the highest ARPU (~\$17,684) on a much smaller base (91 players).

Country Segment

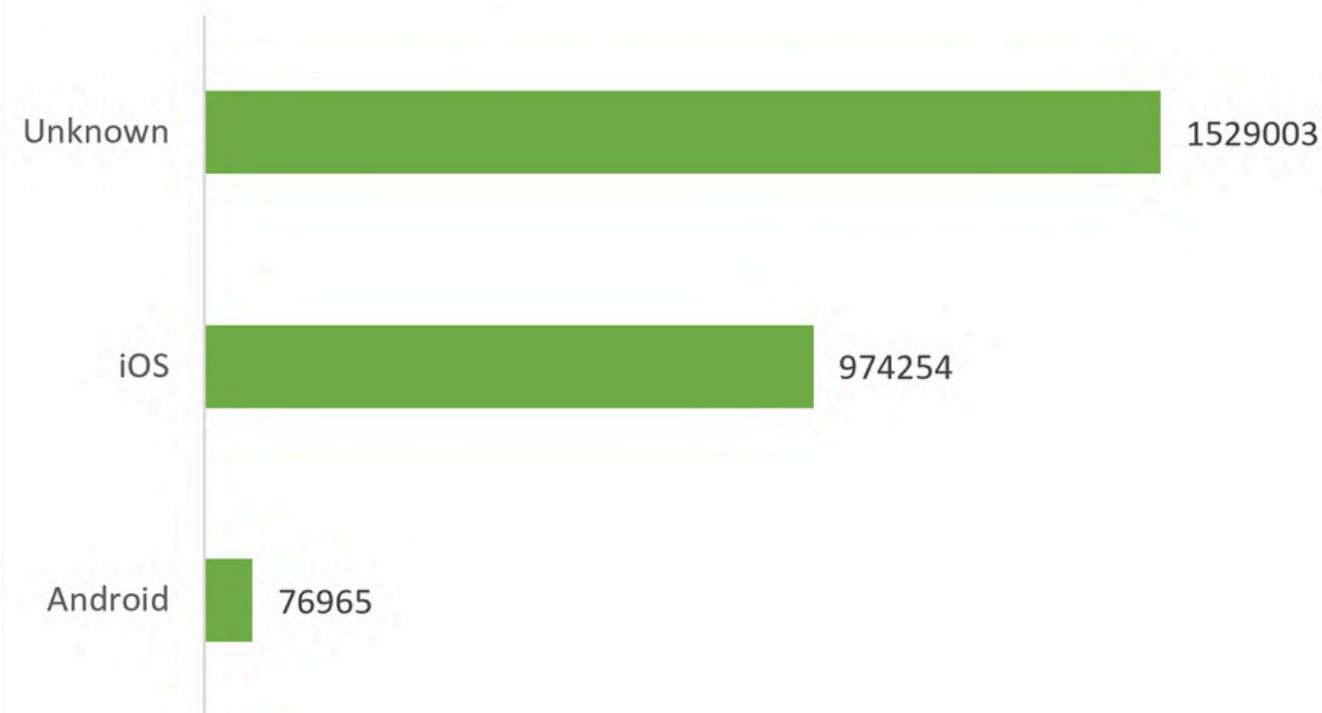


Key Insight #3

Total Revenue



Average Revenue Per Transaction



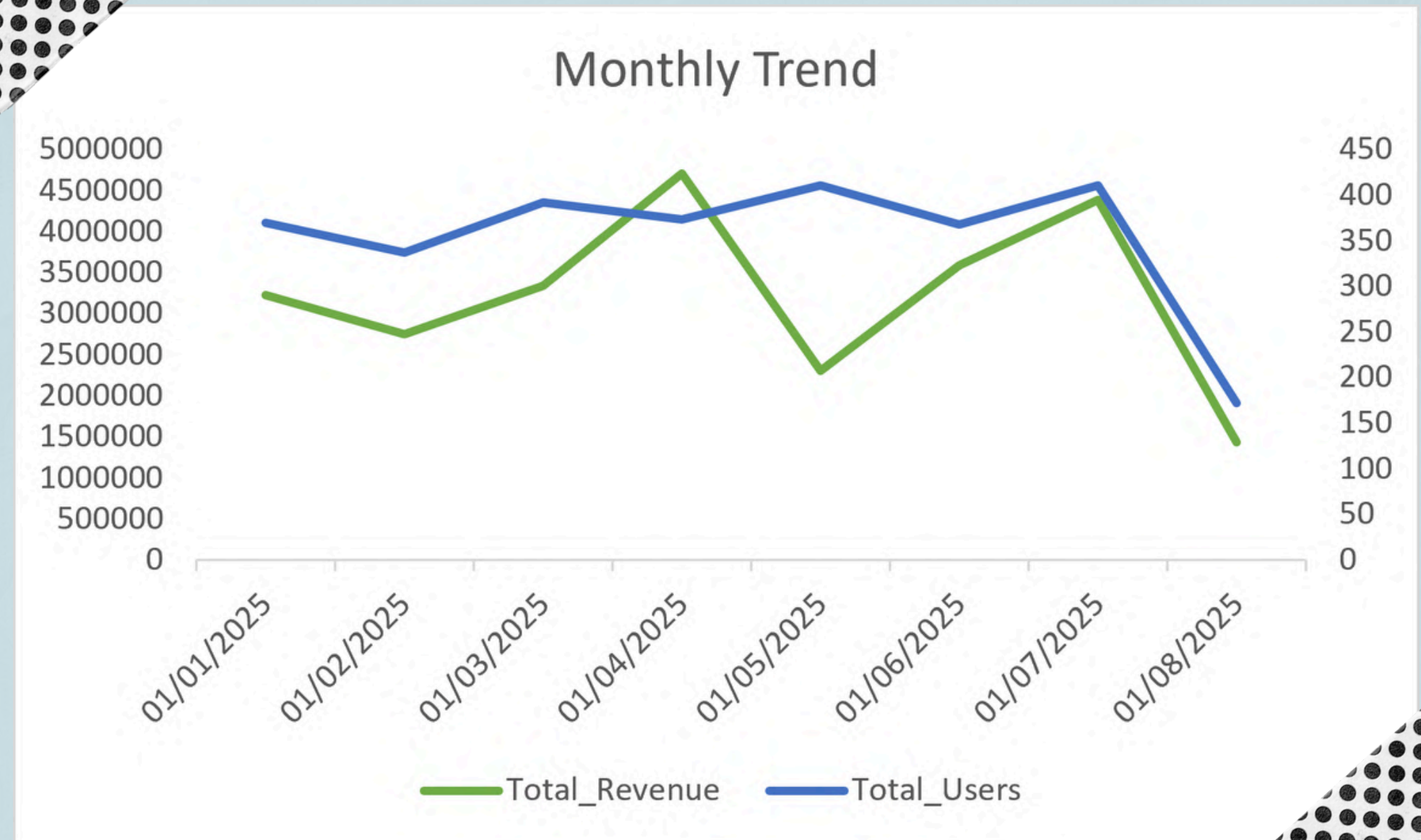
iOS Wins on Value, Android Wins on Scale

Android leads on total revenue (\$13.1M) and player count (1,702). iOS has fewer players (1,203) but a higher average revenue per transaction (\$9,742 vs \$7,696).

Note: the "Unknown" device category shows the highest average transaction value (\$15,290), but only 59 players — too small a sample to act on.

Conclusion & Business Impact

Retention strategy should prioritize the top 2.2% of spenders with VIP treatment — this segment outweighs any single geographic or platform factor in revenue concentration. India and Android should stay volume/ads-focused; Afghanistan-type markets and iOS justify premium-priced content. The same logic applies by genre: Battle Royale and Racing support IAP-heavy design, while Casual and Simulation are better suited to ad monetization.



Thank
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